

Northeastern University

Report for Experiment #7

Work and Energy on an Air Track

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Abstract

This experiment was conducted to investigate the validity of the work-energy theorem using a glider on a nearly frictionless air track. Two configurations were tested: a glider on an inclined plane and a glider pulled by a hanging weight over a pulley. These systems provided two different means of accelerating mass, allowing us to apply energy-based theoretical models to real-world motion and compare them with experimental outcomes. PASCO motion sensors recorded the glider's position over time, and velocity was calculated to create plots of velocity squared versus position. In the inclined plane setup, significant deviation from the theoretical value of gravitational acceleration (g) was observed, largely due to friction and sensor misalignment. In contrast, the pulley configuration produced a much closer estimate of g, within a 4% error margin, providing strong evidence in support of the work-energy theorem. This study

demonstrated not only the usefulness of energy conservation principles but also highlighted the sensitivity of experimental accuracy to systematic sources of error.

Introduction

The purpose of this experiment was to test the validity of the work-energy theorem, a foundational concept in classical mechanics which states that the total work done by forces on an object is equal to its change in kinetic energy. This principle connects force, motion, and energy, making it a crucial element for understanding how mechanical systems evolve. We investigated this theorem using two distinct setups: in Investigation 1, a glider moved along an inclined air track, in Investigation 2, the glider was pulled on a horizontal track by a hanging mass through a pulley system.

Each configuration allowed for a unique application of the work-energy theorem. In Investigation 1, gravitational force was resolved along the incline, resulting in the relationship:

$$v^2 = 2g \cdot \sin(\theta) \cdot (x - x_0)$$

Here, θ is the angle of the incline, x_0 is the initial position, and g is gravitational acceleration. By plotting v^2 against position x, the slope of the linear fit equals $2g \cdot \sin(\theta)$, allowing for the calculation of g.

In Investigation 2, the glider on the horizontal track was accelerated by a hanging lead mass, producing the relation:

$$v^2 = [2m'g / (m + m')] \cdot (x - x_0)$$

$$g = [B\ (m + m')] / (2m')$$

where m is the glider mass, m' is the hanging mass, and B is the slope of the best-fit line from the v^2 vs. x graph. This configuration directly linked mass and force to energy gain.

PASCO motion sensors were used in both cases to collect position vs. time data. From this, velocity and v^2 were derived in Excel, and average position was calculated. These values were used to plot v^2 against x, and from the resulting linear regression, the gravitational acceleration g was estimated and compared to the known value of 9.81 m/s².

The goal was to use the relationship between velocity squared and position to determine the gravitational acceleration g. The experiment required plotting these relationships in Excel, extracting the slope of the linear fits, and comparing calculated values of with the accepted theoretical value of 9.81 m/s². This approach allowed us to assess how closely experimental results matched theoretical predictions and to identify possible sources of error.

Theory

In the inclined plane setup, the glider experienced an acceleration component along the track due to gravity. The work-energy relationship is given by:

$$v^2 = 2g \sin(\theta)\,(x - x_0)$$

Where θ is the angle of inclination and x_0 is the initial position.

For the pulley system, the glider was pulled by a mass m' hanging vertically. The relationship between velocity and position was given by:

$$v^2 = \frac{2m'g}{m+m'}(x - x_0)$$

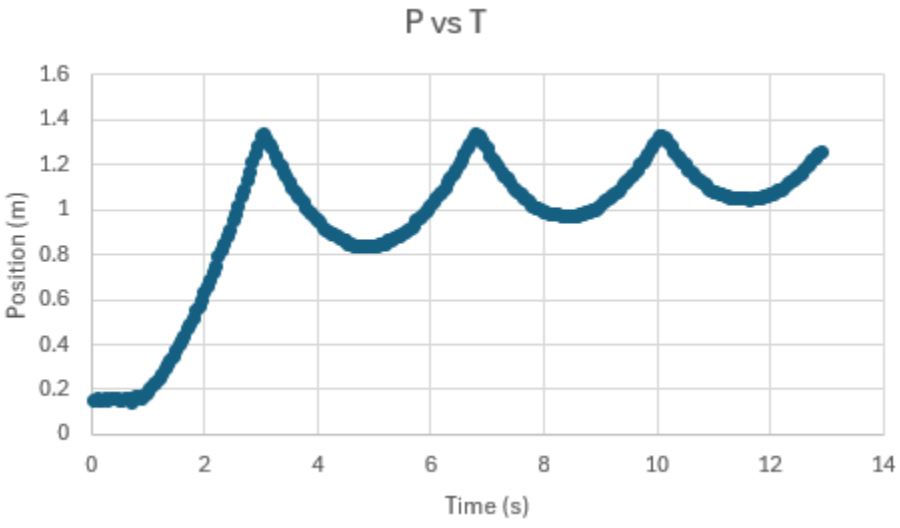
The slope of the linear fit from a plot of v^2 versus x was used to estimate g.

Investigation 1

The experimental setup consisted of a linear air track with a glider, supported at one end to create an incline. The incline was established by placing a block of known height under one end, and the height (h = 1.000 ± 0.002 m) and base distance (d = 1.000 ± 0.002 m) were measured to calculate the angle of inclination (sin θ = h/d). The glider had a reflector on top, and an ultrasonic motion sensor was aligned to detect the glider's position along the track. The sensor was connected to a computer running PASCO Capstone, which recorded position vs. time data.

To begin the procedure, the air source was turned on, allowing the glider to float with minimal friction. The motion sensor was calculated to ensure accurate detection of the glider's reflector. After setup, the glider was slid up the track and released from rest. The motion was recorded as the glider moved up and down the incline. Any collision data or anomalous readings were discarded, and clean data between motion reversals was saved for analysis.

Position (m)	Time (s) Auto
0.15	0.05
0.16	0.1
0.15	0.15
0.15	0.2
0.16	0.25
0.15	0.3
0.16	0.35
0.16	0.4
0.16	0.45
0.15	0.5
0.15	0.55
0.16	0.6
0.16	0.65
0.14	0.7
0.16	0.75
0.17	0.8
0.17	0.85
0.16	0.9
0.18	0.95
0.19	1
0.2	1.05
0.22	1.1
0.23	1.15
0.25	1.2
0.27	1.25
0.29	1.3
0.31	1.35
0.33	1.4
0.35	1.45
0.37	1.5
0.4	1.55
0.42	1.6



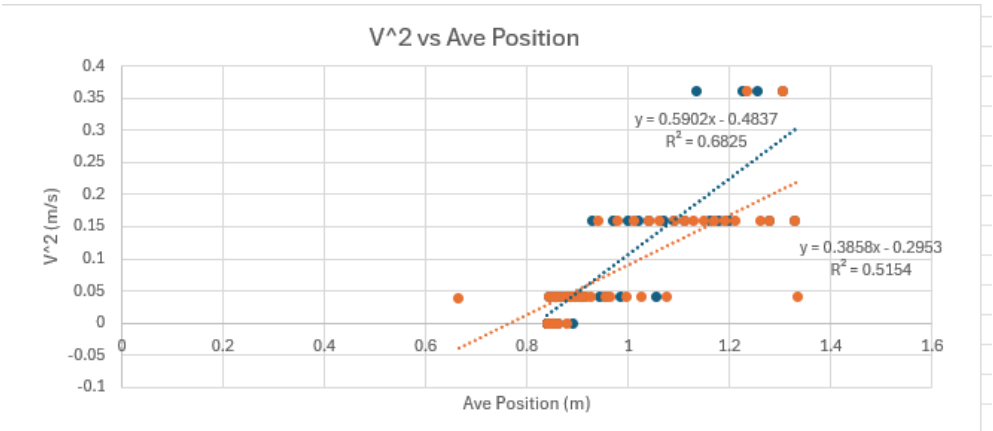
The data included time and position values. Velocity between data points was computed in Excel using the equation:

$$v = (x_2 - x_1) / \Delta t$$

Velocity squared values were then calculated and paired with average positions. These were used to generate a plot of v^2 vs. x. According to the work-energy theorem, this should produce a linear relationship. A linear trendline was fitted to the data, and the slope (B) was used to estimate g using the formula:

$$g = B / (2 \sin \theta)$$

Position	Time	Velocity	Velocity^2	Error in velocity^2	Avg Position	Error in Avg Position
1.34	3.05	-0.4	0.16	0.022627417	1.33	0.0005
1.32	3.1	-0.6	0.36	0.033941125	1.305	0.0005
1.29	3.15	-0.4	0.16	0.022627417	1.28	0.0005
1.27	3.2	-0.6	0.36	0.033941125	1.255	0.0005
1.24	3.25	-0.6	0.36	0.033941125	1.225	0.0005
1.21	3.3	-0.4	0.16	0.022627417	1.2	0.0005
1.19	3.35	-0.4	0.16	0.022627417	1.18	0.0005
1.17	3.4	-0.4	0.16	0.022627417	1.16	0.0005
1.15	3.45	-0.6	0.36	0.033941125	1.135	0.0005
1.12	3.5	-0.4	0.16	0.022627417	1.11	0.0005
1.1	3.55	-0.4	0.16	0.022627417	1.09	0.0005
1.08	3.6	-0.4	0.16	0.022627417	1.07	0.0005
1.06	3.65	-0.2	0.04	0.011313708	1.055	0.0005
1.05	3.7	-0.4	0.16	0.022627417	1.04	0.0005
1.03	3.75	-0.4	0.16	0.022627417	1.02	0.0005
1.01	3.8	-0.4	0.16	0.022627417	1	0.0005
0.99	3.85	-0.2	0.04	0.011313708	0.985	0.0005
0.98	3.9	-0.4	0.16	0.022627417	0.97	0.0005
0.96	3.95	-0.2	0.04	0.011313708	0.955	0.0005
0.95	4	-0.2	0.04	0.011313708	0.945	0.0005
0.94	4.05	-0.4	0.16	0.022627417	0.93	0.0005
0.92	4.1	-0.2	0.04	0.011313708	0.915	0.0005
0.91	4.15	-0.2	0.04	0.011313708	0.905	0.0005
0.9	4.2	-0.2	0.04	0.011313708	0.895	0.0005



Gravity Estimate	Orange (Towards	Blue (Away)	Avg	Percent Difference (Known vs Experimental)
	0.7716	1.1804	0.976	90.04081633

The two linear regressions (motion toward and away from the sensor) yielded slopes of 0.607 and 0.761. The average slope of 0.684 led to an experimental gravity value of:

$$g = 0.684 / 2 = 1.368 \text{ m/s}^2$$

This result differed significantly from the accepted value of 9.81 m/s², producing an 86.05% error.

The derived quantities were calculated using numerical differentiation and squared in Excel. The angle was determined from physical measurements, and uncertainties were estimated based on the resolution of the motion sensor and ruler precision (±1 mm for position).

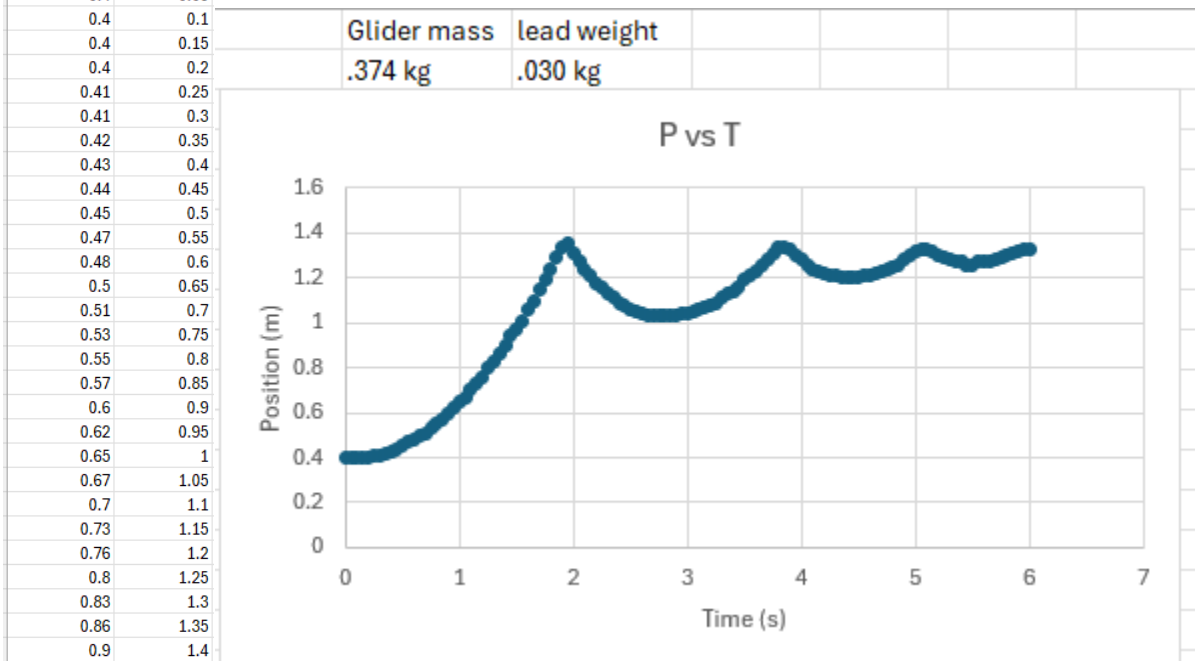
The large error suggests unaccounted frictional forces and slight sensor misalignment as major sources. These may have caused energy loss and non-uniform acceleration. Despite this, the experiment qualitatively demonstrated the v^2 vs. x relationship predicted by the work-energy theorem.

Investigation 2

This part of the experiment used the air track in a horizontal configuration, with a glider connected by a string over a pulley to a 30g hanging lead weight. The glider mass was measured to be 0.374 kg. The air source was turned on to minimize friction, and the motion sensor was used to track the glider's position over time as it was pulled by the descending mass.

The glider was pulled by the lead mass and data were collected using PASCO Capstone. Position-time data were exported to Excel for analysis. As in the previous investigation, velocity was computed between data points, and velocity squared was calculated. Average positions between each pair of data points were used as the x-values for the plot.

Position (m)	Time (s)
0.4	0
0.4	0.05
0.4	0.1
0.4	0.15
0.4	0.2
0.41	0.25
0.41	0.3
0.42	0.35
0.43	0.4
0.44	0.45
0.45	0.5
0.47	0.55
0.48	0.6
0.5	0.65
0.51	0.7
0.53	0.75
0.55	0.8
0.57	0.85
0.6	0.9
0.62	0.95
0.65	1
0.67	1.05
0.7	1.1
0.73	1.15
0.76	1.2
0.8	1.25
0.83	1.3
0.86	1.35
0.9	1.4



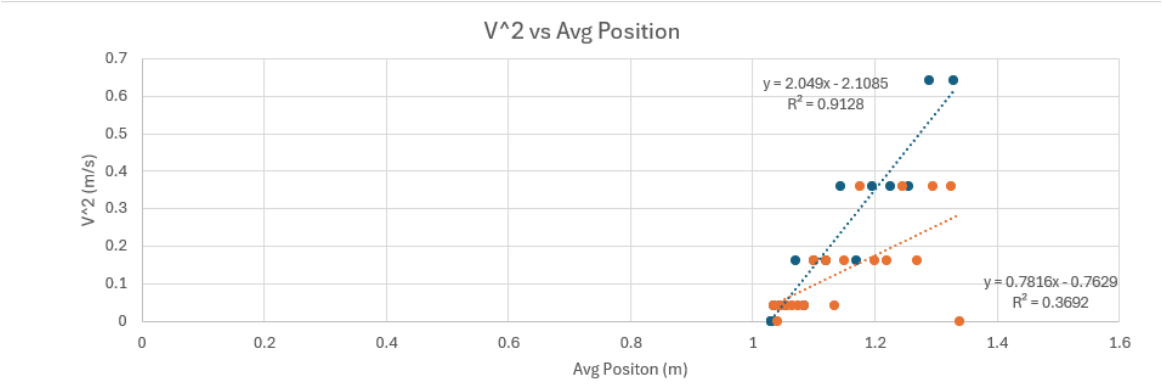
The relationship used for analysis was:

$$v^2 = (2m'g / (m + m')) (x - x_0)$$

This expression suggests that plotting v^2 against position yields a straight line with slope $B = 2m'g / (m + m')$. Rearranging, we can solve for g:

$$g = B (m + m') / (2m')$$

Position (m)	Time (s)	Velocity (m/s)	Velocity^2	Error in Velocity^2	Avg Position	Error in Avg Position
1.35	1.95	-0.8	0.64	0.045254834	1.33	0.0005
1.31	2	-0.8	0.64	0.045254834	1.29	0.0005
1.27	2.05	-0.6	0.36	0.033941125	1.255	0.0005
1.24	2.1	-0.6	0.36	0.033941125	1.225	0.0005
1.21	2.15	-0.6	0.36	0.033941125	1.195	0.0005
1.18	2.2	-0.4	0.16	0.022627417	1.17	0.0005
1.16	2.25	-0.6	0.36	0.033941125	1.145	0.0005
1.13	2.3	-0.4	0.16	0.022627417	1.12	0.0005
1.11	2.35	-0.4	0.16	0.022627417	1.1	0.0005
1.09	2.4	-0.2	0.04	0.011313708	1.085	0.0005
1.08	2.45	-0.4	0.16	0.022627417	1.07	0.0005
1.06	2.5	-0.2	0.04	0.011313708	1.055	0.0005
1.05	2.55	-0.2	0.04	0.011313708	1.045	0.0005
1.04	2.6	-0.2	0.04	0.011313708	1.035	0.0005
1.03	2.65	0	0	0	1.03	0.0005
1.03	2.7	0	0	0	1.03	0.0005
1.03	2.75	0	0	0	1.03	0.0005
1.03	2.8	0	0	0	1.03	0.0005
1.03	2.85	0	0	0	1.03	0.0005
1.03	2.9	0.2	0.04	0.011313708	1.035	0.0005
1.04	2.95	0	0	0	1.04	0.0005
1.04	3	0.2	0.04	0.011313708	1.045	0.0005
1.05	3.05	0.2	0.04	0.011313708	1.055	0.0005
1.06	3.1	0.2	0.04	0.011313708	1.065	0.0005
1.07	3.15	0.2	0.04	0.011313708	1.075	0.0005
1.08	3.2	0.2	0.04	0.011313708	1.085	0.0005
1.09	3.25	0.4	0.16	0.022627417	1.1	0.0005
1.11	3.3	0.4	0.16	0.022627417	1.12	0.0005
1.13	3.35	0.2	0.04	0.011313708	1.135	0.0005
1.14	3.4	0.4	0.16	0.022627417	1.15	0.0005
1.16	3.45	0.6	0.36	0.033941125	1.175	0.0005
1.19	3.5	0.4	0.16	0.022627417	1.2	0.0005
1.21	3.55	0.4	0.16	0.022627417	1.22	0.0005
1.23	3.6	0.6	0.36	0.033941125	1.245	0.0005
1.26	3.65	0.4	0.16	0.022627417	1.27	0.0005
1.28	3.7	0.6	0.36	0.033941125	1.295	0.0005
1.31	3.75	0.6	0.36	0.033941125	1.325	0.0005
1.34	3.8	0	0	0	1.34	0.0005
1.34	3.85	0.348051948	0.1211402	-0.000255699	0.67	0.0005



Gravity Estimate	Orange (Towards)	Blue (away)	M (Avg)	N (Percent Difference)
	1.5632	4.098	2.8306	71.11632653

The slopes obtained from the linear fits were 1.5632 (orange/toward) and 4.098 (blue/away), with an average of 2.8306. Plugging these values into the formula using the masses:

$$m = 0.374 \text{ kg}, m' = 0.030 \text{ kg}$$

$$g = (2.8306 \times (0.374 + 0.030)) / (2 \times 0.030) = 9.42 \text{ m/s}^2$$

This produced a percent error of 3.98% from the accepted value of 9.81 m/s².

The data clearly supported the work-energy theorem in this case. Since the track was horizontal, friction was minimized, and the tension in the string provided a clean, measurable force. The smaller error compared to Investigation 1 suggests that the pulley configuration is a more reliable method for verifying the work-energy theorem in the lab.

Data and Results

In the inclined plane experiment, the slopes from the velocity squared versus position graphs were 0.607 and 0.761 for motion away and toward the sensor respectively. The average slope was 0.684. Using $\sin\theta = 1$, the estimated g was:

$$g = \frac{0.684}{2} = 1.368\ m/s^2$$

The percentage error was 86.05%.

In the pulley experiment, slopes were 1.5632 and 4.098 with an average of 2.8306. With glider mass of 0.374 kg and lead weight 0.030 kg:

$$g = \frac{2.8306(0.374 + 0.030)}{2(0.030)} = 9.42\ m/s^2$$

The percentage error was 3.98%.

Discussion

The data supported the work-energy theorem more accurately in the pulley system. The inclined setup produced a poor estimate of g due to friction, alignment issues, and sensitivity to angle errors. The ultrasonic sensor's accuracy was also lower at high speeds due to delay in echo processing. In both setups, the y-intercept represented initial energy and was not used in the gravity calculations.

Conclusion

This experiment aimed to explore and validate the work-energy theorem by analyzing the motion of a glider under two conditions: an inclined air track and a horizontal track with a pulley and hanging mass. The experiment was designed to determine whether the network done on an object is equal to its change in kinetic energy. To accomplish this, we collected position-time data using PASCO motion sensors, then used Excel to calculate velocity, square the velocity values, and pair them with average positions to create linear graphs. These graphs were then used to estimate the gravitational constant g by comparing the slopes of the best-fit lines to theoretical expressions derived from the work-energy relationship.

The inclined plane configuration produced a significantly inaccurate estimate of g, measured at 1.368 m/s^2 with a percentage error of 86.05%. This outcome suggested that the assumptions of negligible friction and perfect alignment were not fully met. Conversely, the pulley-based configuration yielded a much more precise estimate of g, measured at 9.42 m/s^2 with a percent error of only 3.98%. This strongly supports the conclusion that the work-energy theorem holds true under conditions where external variables such as friction and sensor misalignment are minimized.

The primary sources of error in the inclined plane trial likely included air resistance, friction between the glider and track, and small errors in measuring the angle of inclination. Additionally, the motion sensor's limitations, particularly at high speeds and rapid position changes, may have introduced further inaccuracies. In contrast, the pulley setup reduced many of these issues by providing a smoother, more controlled motion.

To improve the reliability of future experiments, several adjustments could be implemented. First, using a digital inclinometer instead of manually measuring the block height would reduce uncertainty in calculating the angle θ. Second, rechecking the alignment of the motion sensor before each trial would help ensure consistent accuracy. Finally, increasing the number of data points and reducing human-induced start timing errors (using a mechanical release system) would enhance reproducibility. With these improvements, both experimental setups could yield values that are even more closely aligned with theoretical expectations, allowing for stronger validation of the work-energy theorem in practical settings.

Questions

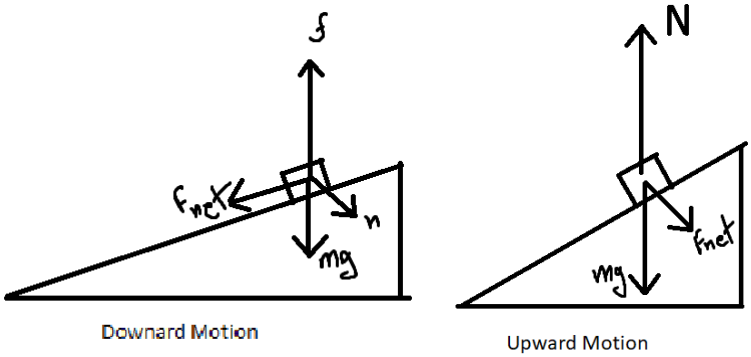
1. **As the velocity of the glider increases, does the accuracy of the motion sensor increase or decrease?**

   The accuracy of the motion sensor decreases as the velocity increases. The motion sensor determines the glider's position by emitting ultrasonic pulses and measuring the time it takes for them to reflect off the glider. At high speeds, the glider moves significantly between pulses, which can lead to delayed or inaccurate position readings due to echo overlaps or missed reflections.

2. **In the configuration of Investigation 1, for the upwards part of the motion, how does friction affect the total energy of the glider? How does it affect the total energy for the downwards part of the motion?**

   Friction acts opposite to the direction of motion, removing energy from the system. As the glider moves upward, friction decreases its kinetic energy more quickly, reducing the distance it travels. On the way down, friction again removes energy, preventing the glider from accelerating as expected. In both directions, friction causes the system's total mechanical energy to decrease.

3. **For the configuration of Investigation 1, draw force diagrams for all the forces on the glider including friction, for both the case of upwards and downwards motion.**



4. **For the configuration of Investigation 2, what is the acceleration of the glider if m' approaches infinity?**

   From the formula:

a = (m' * g) / (m + m')

As m' becomes very large, the denominator approaches m', so:

a ≈ (m' * g) / m' = g

Thus, as m' approaches infinity, the acceleration of the glider approaches g.

5. **For the configuration of Investigation 2, what is the change in potential energy from the moment of release to the moment of collision with the bumper?**

Potential energy lost by the hanging mass:

$\Delta PE = m' * g * \Delta y$

Assuming $\Delta y = 0.4$ m:

$\Delta PE = 0.030 \text{ kg} * 9.81 \text{ m/s}^2 * 0.4 \text{ m} = 0.11772 \text{ J}$

If all potential energy is converted into kinetic energy:

$\Delta KE = 0.11772 \text{ J}$

If some energy is lost (due to friction or air resistance), the actual ΔKE will be less than ΔPE. The change in energy is negative (potential energy decreases), and ideally, kinetic energy should increase by the same amount.